to themselves. Defending against a DoS attack on the other hand can require significant resources, costing time and money. But there are several basic steps you can take to reduce the opportunity for DoS attacks.

Many DoS attacks are caused using unauthenticated requests. Imagine you're driving down a street and suddenly the traffic lights at the pedestrian crossing up ahead turn red, bringing the traffic to a halt, but nobody is waiting to cross. Unknown to you, a few moments earlier somebody pressed the button on the crossing and then ran off before anyone could see them. This is an example of a mostly harmless DoS attack. By pressing the button when they did not need to cross, they managed to bring the traffic to a stop (denying use of the road temporarily) at no cost to themselves. By running away before the lights changed, they reduced the risk of being recognized. Many online DoS attacks operate under the same assumption. An attacker does not want to authenticate to your API, because this may slow them down, reducing the number of malicious requests they can make, and runs the risk of them being identified. In many cases authentication isn't needed to get the server to use resources. One simple way to prevent these kinds of attacks is therefore to never let unauthenticated requests consume resources on your servers. Authentication is covered in section 3.4 and should be applied immediately after rate-limiting before any other processing.

> **TIP** Never allow unauthenticated request to consume significant resources on your server.